

Module 06

Dockerfile Advance

DevOps end to end services & solutions

Agenda

- > Dockerfile Advance
- > Docker Advance commands
- > Hands-on

Dockerfile – Port set from environment

```
FROM alpine:3.19
RUN apk add --no-cache
python3
WORKDIR /app
COPY index.html .
ENV PORT=5022
EXPOSE 5022

CMD ["sh", "-c", "python3 -m
http.server $PORT"]
```

FROM alpine:3.19

- Uses **Alpine Linux** as the base image
- Very small (~7 MB)
- Good for simple services

RUN apk add --no-cache python3

- Installs **Python 3** inside the image
- --no-cache keeps the image small
- Required to run the HTTP server

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http.server $PORT"]
```

WORKDIR /app

- Sets /app as the **working directory**
- All next commands run from /app
- Creates the directory if it doesn't exist

COPY index.html .

- Copies index.html from your host
- Places it inside /app in the image
- This is what the HTTP server will serve

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CMD ["sh", "-c", "python3 -m
http.server $PORT"]
```

ENV PORT=5022

- Sets an **environment variable**
- Default port the app will listen on
- Can be overridden at runtime:

EXPOSE 5022

- container listens on port **5022**

Dockerfile – Port set from environment

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http.server $PORT"]
```

CMD ["sh", "-c", "python3 -m http.server
\$PORT"]

- Command that runs when the container starts
- Starts Python's built-in HTTP server
- Uses \$PORT from the environment

Docker Build and Run

Docker top < C ID >

It shows you **what processes are actually running inside a container**

Docker stats < C ID >

live performance dashboard for containers
It shows **real-time resource usage** — CPU, memory, network, and disk I/O

Docker Build and Run

Docker inspect < C ID >

deep-dive X-ray of Docker

It shows **everything Docker knows** about an image or container — config, networking, mounts, env vars, ports, PID, etc

Docker logs –follow < C ID >

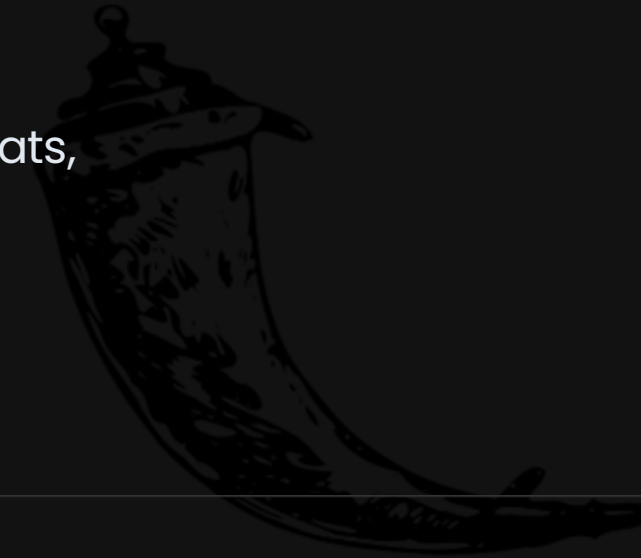
watch a container's output live

Like less

Docker logs < C ID > | less

Hands-On – Docker Argument port and Jenkins

- >
- > • Go to Repo – Docker Advance commands `docker top, stats,`
- > `inspect, etc`
- >



Flask

Questions?

Thank you for participating.

Feel free to reach out:
<https://dinghy-e2e.com>

